

Overview of the Multimedia Grand Challenges 2022

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ABSTRACT

The Multimedia Grand Challenge track was first presented as part of ACM Multimedia 2009 and has established itself as a prestigious competition in the multimedia community. The purpose of the Multimedia Grand Challenges is to engage the multimedia research community by establishing well-defined and objectively judged challenge problems intended to exercise the state-of-the-art methods and inspire future research directions. The key criteria for Grand Challenges are that they should be useful, interesting, and their solution should involve a series of research tasks over a long period of time, with pointers towards longer-term research. The 2022 edition of ACM Multimedia hosted 10 Grand Challenges covering all aspects of multimedia computing, from delivery systems to video retrieval, from video generation to audio recognition.

CCS CONCEPTS

• **Information systems** → **Multimedia information systems; Multimedia and multimodal retrieval**; • **Human-centered computing** → **Interactive systems and tools**.

KEYWORDS

challenges, multimedia

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1 INTRODUCTION

ACM Multimedia is the premier international conference in the area of multimedia within the field of computer science. Multimedia research is rich and diverse, focusing on integration of the multiple perspectives offered by different digital modalities including images, text, video, music, sensor data, spoken audio.

Since 2009, Multimedia features a Grand Challenge track, allowing researchers and practitioners to organize interesting, objectively-judged competitions focused on multimedia topics. The Multimedia Grand Challenges reflect the breath and depth of the multimedia

field, offering engaging venues for researchers to experiment with existing and novel methods on specific applied problems.

Similar to previous editions, this year prospective challenges organizers responded to an initial call for Multimedia Grand Challenge proposals. Proposals had to feature a description of the challenge, an overview of the benchmark datasets provided, and the set of evaluation metrics. This year, we received a record number of 14 grand challenge proposals. We selected the top 10 challenges according to data availability, objectiveness of the metrics proposed, interestingness of the challenge to the multimedia community, and feasibility of the tasks. The organizers of the 10 accepted challenges then individually managed tasks, competition guidelines, solution submissions and engagement with participants. Participants were also invited to send papers to be peer-reviewed and possibly published within the ACM Multimedia 2022 proceedings. The top-ranked participants for each challenge will present live at a plenary Grand Challenge session in the main conference.

2 OVERVIEW OF THE 2022 CHALLENGES

In this Section, we summarize the topics, submission formats, and findings for the 10 competitions selected for the 2022 edition of the MM Grand Challenge. Most of the content of each subsection comes directly from the challenge organizers, some giving more detailed information than others. Several challenges also have specific overview papers: Short Video Streaming, Deep Video Understanding, Computational Paralinguistics, Facial Micro-Expression and MultiMediate.

2.1 Social Media Prediction (SMP) Challenge

The ubiquity of social media (e.g. Facebook, Twitter, Flickr, YouTube, etc.) provides a crucial pathway for learning and even forecasting about user preferences or behaviors the real-world. The SMP Challenge (Social Media Prediction Challenge) [16, 18, 19] explores novel challenge tasks and novel methods based on numerous resources on social media and meaningfully improving people's social lives and business scenarios. The activity hosted for 5 years and already attracted over 120 teams. The challenge provided the SMPD (Social Media Prediction Dataset), a large-scale, multimodal social media dataset with half million of online posts, temporal popularity information, and their metadata from about 70K users. So far the researches of social media prediction covered in several significant areas of multimedia and artificial intelligence, and closely integrated with computer vision, machine learning, natural language and human-centered interaction. The researches benefit to many applications of online advertising, social marketing, personalized recommendation, demand forecasting, etc.

The task of the SMP Challenge Social Media Popularity Prediction (SMPP) [17, 19] aims at predicting the impact of online post

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sharing on social media. It formulates the task at a temporal prediction problem [18] and the popularity by online attentions based on various user interactions (e.g. clicks, visits, reviews). Existing work in the early stage mainly pays attention to the following challenges: feature selection [3, 8, 10, 12], single network architecture [5], and model fusion [6]. These challenges were shared with the primary issues of mainstream machine learning or deep learning tasks, and the proposed methods ignored high-level challenges of social media prediction. Several new ideas were proposed motivated by semantic relevance [4, 7], embedding representation [4, 5, 11], user-post interactions [3], and aligned visual-textual information. The proposed methods are insightful and started to uncover the inherent characteristics of the perspectives of high-level correlations. They involved the attention, recurrent, or sliding window mechanisms in their models [4, 9, 14, 20] and achieved efficient improvements. This year, the submissions illustrate several new directions compared with the previous years of studies. The new work this year is not following the conventional paradigm, they are exploring contrastive learning, image-text representation alignment, self-attention, cross-attention, or transformer architecture. The average overall performance improvements of the SRC and MAE are about 13.9% and 0.17. The result of the champion this year also achieved the best and outperforms the first place teams in previous years of the SMP Challenge.

2.2 Short Video Streaming Challenge

This challenge focuses on optimizing the delivery of short videos. Participants are invited to submit algorithms that reduce bandwidth usage without impacting the quality of experience. This year, 27 teams completed a challenge submission. Each team had 1-3 members. Most top teams used heuristic algorithms and 2 adopt learning-based algorithms. All submitted heuristic-based solutions utilized a bandwidth estimation scheme to assist in decision-making. Submitted learning-based solutions used actor-critic frameworks in reinforcement learning and learned from past streaming experiences. Among the top 10 teams, the two learning-based algorithms placed 8th and 10th: this suggests that the performance of learning-based schemes may be affected by the lack of generalization when faced with various combinations of different network conditions, video sequences, and user behaviors.

2.3 Deep Video Understanding Challenge

The DVU Challenge invites solutions to address how an automatic system understands and comprehend a full movie in terms of entities (characters, locations, concepts, relationships, interactions, sentiments, etc.). The challenge provides a large dataset of movies annotated at the movie-level and scene-level. To foster participation, this year's challenge was organized into 4 Tracks - 2 over the movie level and 2 over the scene-level. The 2022 DVU edition saw 6 participating teams for a total 29 individuals. Good solutions to this challenge were truly multi-modal, making full use of video, text, and audio features. No single one of these modalities could be reliably said to be more important than the others. Results of the 2022 edition show an improvement over previous years of the challenge, suggesting that our community is making progress on Deep Video Understanding methods.

2.4 Computational Paralinguistics ChallengeE

The Computational Paralinguistics ChallengeE (ComParE) series is an open Challenge in the field of Computational Paralinguistics, dealing with states and traits of individuals as manifested in their speech and further signals' properties. This year, the challenge hosted four tracks: (1) Vocalisations Sub-Challenge, (2) Stuttering Sub-Challenge, (3) Activity Sub-Challenge, and (4) Mosquitos Sub-Challenge. 68 solutions were submitted for this year's edition of the challenge. Most participants used wav2vec2-based systems [2] successfully, alone or by ensembling them with other methods. Additionally, data augmentation techniques, domain classifiers and pitch conversion were employed. Transformer-based models were used for the first time for the detection of mosquitoes.

2.5 Detecting CheapFakes

Cheapfake is a recently coined term that encompasses non-AI ("cheap") manipulations of multimedia content, created using software for image/video editing, or by simply sharing the media alongside misleading claims. This challenge focuses on detecting the misuse of real photographs with conflicting image captions in news items, based on the recently compiled COSMOS dataset [1]. This year's edition hosted 2 tasks, and received 5 submissions. One of the main challenges for participants is to propose methods that outperform the COSMOS baseline approach, a deep learning-based model exploiting different sources of information. This year, participants focused on finding hidden clues rather than improving existing network-bbased models, finding interesting hints e.g. from the sentiment of the caption caused by negative words. This makes the proposed approaches more robust than existing methods.

2.6 Facial Micro-Expression Grand Challenge

This challenge focuses on the detection of facial micro-expressions (MEs), involuntary movements of the face that occur spontaneously when a person experiences an emotion but attempts to suppress or repress the facial expression. This year's challenge was organized into 2 tracks: the ME generation task and the ME and the macro-expression spotting task. In total, 11 teams participated to the MEGC2022 challenge, with 3 teams for the generation task and 8 teams for the spotting task. Submissions providing special treatments for glasses, such as the face parsing model, improved the quality of the generated videos. For the spotting task, most of the participants chose to use optical flow as a feature to learn the movement information of facial expressions. Similar to last year, the method of difference comparison after handcraft feature extraction still outperforms the spotting method based on deep learning, due to the limited size of the dataset.

2.7 GigaTracking: When Multi-Object Tracking meets Gigapixel Videography

When it comes to gigapixel-level images/videos, state-of-the-art tracking algorithms perform unsatisfactorily due to traits like wide field of view, high resolution, and intricate object interactions. The GigaTracking challenge aims to find new solutions toward accurate Multi-Object Tracking for gigapixel-level videos. In particular, the participants are encouraged to investigate the questions, including but not limited to novel representations, network architecture, and

paradigm, so as to explore advanced solutions for contributing the communities.

2.8 MultiMediate challenge

The MultiMediate challenge is designed to work towards the vision of effective artificial mediators to support group conversations, by facilitating and measuring progress on key group behaviour sensing and analysis tasks. This year, the challenge focused on backchannel detection, agreement estimation from backchannels, eye contact detection and next speaker prediction. Two teams successfully submitted test predictions, achieving a new state-of-the-art on the eye contact detection subchallenge based on an end-to-end trainable network, and improved results for backchannel detection and agreement estimation based on graph neural networks.

2.9 Conversational Head Generation

Conversational head generation consists in synthesizing the head dynamics during a conversation. This challenge is based on ViCo dataset [21], the first video corpus featuring conversations between real humans, and invites submissions around talking head video generation (speaker), and responsive listening head video generation (listener). This year, the challenge received a total of 125 submissions for the two tasks. Participants found that extracting semantic information from audio plays an important role in the listening head generation track and the long-range dependencies cross time during video generation can improve the vividness for both tasks. Moreover, prior knowledge such as GFP-GAN [15] can make the generation more high-fidelity and high-resolution.

2.10 Pre-training for Video Understanding Challenge

The goal of this challenge is to offer a fertile ground for designing pre-training techniques that facilitate a series of video understanding downstream tasks. This year's challenge was organized into two tracks: Pre-training for Video Captioning and Pre-training for Video Categorization. Submission were based on two large-scale video pre-training datasets, i.e., Auto-captions on GIF (ACTION) [13] and a Weakly-Supervised dataset. The challenge received 10 submissions in total. For both tracks, the top-performing teams adopted several common components in their systems, e.g., model ensemble and fusion of different image/video backbones. More interestingly, the best-performing team in the second track leverages the ensembling over multiple temporal resolutions, which is an effective way to boost the accuracy of single model.

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